



Ming-Hsuan Yen

**Natural Catastrophe & Seismic Hazard Modelling | Geospatial Data Scientist |
PhD Geophysics**

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sponsorship required

Professional Summary

Geophysicist (PhD) with 9 years of experience modelling earthquake hazard from real-world data across Taiwan, Japan, New Zealand and Europe. Builds Python data pipelines and statistical and physics-based models that turn large, heterogeneous seismic and geospatial datasets into inputs for hazard and risk assessment – the building blocks of catastrophe modelling. Industry exposure at Aon Impact Forecasting. Experienced in communicating results to government agencies and policy audiences.

Core Competencies

- AI-assisted workflow/ Agentic AI
- Data pipeline development & automation
- Statistical & numerical modeling
- Time-series data analysis
- Risk quantification & uncertainty analysis
- Python / scientific computing
- Geospatial data analysis (GIS)
- Cross-functional & international collaboration
- Technical reporting & stakeholder communication
- Scientific & data quality control

Professional Experience

Data Science Projects – Career Transition

mingsLab, Potsdam, Germany (Apr 2026 – Present)

Career transition from research to industry: portfolio projects in seismic hazard data science (mingslab.com).

- Near-Fault Pulse Database (live): automated Python pipeline that pulls earthquake records from global archives, detects damaging velocity pulses and updates an interactive map every 6 hours; reproduces 25 of 27 published results, backed by 70 automated tests.
- Built with AI coding agents; AI agents fetch the data and run quality checks on a scheduled pipeline.

Guest Researcher (part-time)

GFZ Helmholtz Centre for Geosciences, Potsdam, Germany (Apr 2025 – Mar 2026)

- Co-authored peer-reviewed publications on earthquake ground motion and seismic hazard; in parallel completed an intensive AI Software Development course (WBS Coding School, 2026).

Scientific Researcher

Seismic Hazard and Risk Dynamics, GFZ Helmholtz Centre, Potsdam, Germany (Nov 2019 – Mar 2025)

EU Horizon 2020 Marie Skłodowska-Curie project URBASIS (urban engineering seismology): earthquake hazard modelling with large-scale seismic data.

- Built a continental-scale data pipeline (Python, SQL) that reduced ~52 million raw seismic waveforms from ~1,600 stations in Germany, Italy, France and Poland to 220,000 quality-controlled spectra and a catalogue of 6,135 characterised earthquakes.
- Developed and validated physics-based and statistical models of earthquake sources and ground-motion scaling; results calibrate ground-motion simulations for European seismic hazard assessment.
- Designed signal-processing workflows for multi-network strong-motion data (Taiwan, Japan, New Zealand, Türkiye, Europe); showed that magnitude-based models miss strong site-to-site variability in damaging near-fault ground motion.

- Communicated hazard modelling results to non-specialist and policy audiences (reports, talks).
- Published 7 peer-reviewed papers, 4 as first author (selected list below).

Research Intern

Impact Forecasting, Aon, Prague, Czechia (Apr 2022 – May 2022)

- Compiled, cleaned and harmonised earthquake and site-condition data for Italy and Switzerland as input to a site-model update for Impact Forecasting's seismic hazard and risk models.
- Worked with model developers to understand the end-to-end catastrophe-modelling workflow, from hazard and site inputs to Aon's internal modelling systems.

Research Assistant

Earthquake Disaster & Risk Evaluation and Management Center, National Central University, Taiwan (Aug 2016 – Sep 2019)

National programmes: Taiwan Earthquake Model; spent nuclear fuel final disposal program.

- Delivered seismic site risk assessments to government agencies for a spent nuclear fuel disposal site, integrating evidence from geologists, seismologists and engineering seismologists.
- Ran broadband ground-motion simulations to estimate earthquake source parameters for the site evaluation, validated against recorded seismic data (published 2021).
- Simulated historical earthquakes for the Taiwan Earthquake Model and supported model/data exchange with teams in Japan and New Zealand.

Education

Dr. rer. nat., Geophysics

Universität Potsdam, Germany (2019 – 2024)

*EU Marie Skłodowska-Curie Fellow (URBASIS). Graduated magna cum laude.

*Thesis: Earthquakes Source Parameters and Near-source Velocity Pulses Variabilities in the Context of Ground Motion Prediction

M.Sc., Geophysics

National Central University, Taiwan (2014 – 2016)

*Thesis: The source rupture analysis and 3D seismic wave simulations of the 1935 Hsinchu-Taichung Earthquake

B.Sc., Earth Science

National Central University, Taiwan (2010 – 2014)

Technical Skills

Programming	Python (pandas, NumPy, SciPy, ObsPy, pytest), SQL, MATLAB, FORTRAN, TypeScript
Tools	OpenQuake, QGIS, Jupyter, Linux/Unix, Git/GitHub, AI coding agents
Languages	English (advanced), German (B1, working towards B2), Mandarin (native)

Selected Publications (Full publication list available upon request.)

Yen et al. (2025) - An analysis of directivity pulses using empirical data and dynamic rupture simulations of the 2023 Kahramanmaraş earthquake doublet. *Earthquake Spectra*. doi: <https://doi.org/10.1177/87552930241305012>

Yen et al. (2024) - Source parameters and scaling relationships of stress drop for shallow crustal seismic events in Western Europe. *Journal of Seismology*. doi: <https://doi.org/10.1007/s10950-023-10188-y>

Yen et al. (2022) - Within- and between-event variabilities of strong-velocity pulses of moderate earthquakes within dense seismic arrays. *Bulletin of the Seismological Society of America*. doi: <https://doi.org/10.1785/0120200376>

Further Training

AI Software Development, WBS Coding School (2026)
IBM Data Science Professional Certificate (2025)